

COST OPTIMIZATION

Cost optimization:

Cost optimization means reducing cloud spend without affecting performance, security, or availability by using the right services, sizes, and pricing models.

Why Cost Optimization Is IMPORTANT (Interview Points)

1 Avoid Wasted Money

- Unused EC2, idle EBS volumes, old snapshots still cost money
- NAT Gateway running 24/7 = hidden high cost

2 Better Business Profit

- Lower infra cost = higher profit margin
- Especially critical for startups & SMEs

3 Scale Without Fear

- Optimized cost allows safe scaling during peak traffic
- Prevents bill shocks at month-end

4 Budget Control & Forecasting

- Predict next month's cost
- Prevent unexpected AWS account suspension

5 Compliance & Management Reporting

- Finance teams need clear cost visibility
- Tag-based reporting helps audits

6 Performance-Cost Balance

- Use right-sized instances instead of over-provisioning

AWS Cost Optimization TOOLS (MOST IMPORTANT)

◆ 1 AWS Cost Explorer

Primary tool for cost analysis

Used for:

- Daily / monthly cost breakdown
- Cost by service, region, account
- Forecast future spending

Example Use Case:

Identify NAT Gateway causing high bill

◆ 2 AWS Budgets

Prevent overspending

Used for:

- Set monthly budget (example: \$50)
- Send email/SNS alert at 80%, 100%

Example Use Case:

Alert before free-tier limit crosses

◆ 3 AWS Trusted Advisor

Cost + security recommendations

Cost checks:

- Idle EC2 instances
- Underutilized EBS volumes
- Unused Load Balancers

Note:

- Full checks require Business / Enterprise support

◆ **Compute Optimizer**

AI-based right-sizing tool

Used for:

- Recommend EC2 size changes
- Identify over-provisioned instances

Example:

Change t3.large → t3.small

◆ **5 Savings Plans**

Commit usage for lower cost

Types:

- Compute Savings Plan
- EC2 Instance Savings Plan

Savings:

- Up to 66% cheaper

◆ **6 Reserved Instances (RI)**

Long-term workloads

Used for:

- Databases (RDS)
- Steady EC2 usage

Commitment:

- 1 year / 3 years

◆ **7 AWS Cost Anomaly Detection**

AI-based alert system

Used for:

- Detect unusual cost spikes automatically

◆ 8 Tagging & Cost Allocation Tags

Track who is spending what

Examples:

Environment = Prod

Project = PaymentApp

Owner = DevTeam

◆ 9 Auto Scaling

Pay only for what you use

Use Case:

- Scale out during peak
- Scale in at night

◆ 10 Spot Instances

Cheapest EC2 option

Used for:

- Batch jobs
- CI/CD
- Non-critical workloads

Common Cost Optimization Actions (Real Life)

Resource	Optimization
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EC2	Right-size / stop unused
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EBS	Delete unattached volumes
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NAT GW	Use VPC endpoint instead
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- S3 Move to IA / Glacier
- RDS Reduce size / reserved instance
- CloudWatch Reduce log retention

Cost Billing Analysing:

1. Cost bill analysis helps identify **where money is being spent** across AWS services and regions.
2. It detects **unused or underutilized resources** like idle EC2, EBS volumes, and snapshots.
3. Analyzing bills prevents **unexpected cost spikes** and budget overruns.
4. It supports **cost optimization decisions** such as right-sizing, Reserved Instances, and Savings Plans.
5. Regular analysis ensures **better financial control, forecasting, and accountability** in cloud usage.

Last month's billing

	Total	June 2025*	July 2025*	August 2025*	September 2025*	October 2025	November 2025
Total costs	\$25.53	\$0.00	\$0.00	\$0.00	\$0.00	\$0.00	\$25.53
VPC	\$11.90	-	-	-	-	\$0.00	\$11.90
EC2-Other	\$9.60	-	-	-	-	\$0.00	\$9.60
Tax	\$3.90	-	-	-	-	\$0.00	\$3.90
EC2-Instances	\$0.13	-	-	-	-	\$0.00	\$0.13
S3	\$0.00	-	-	-	-	\$0.00	\$0.00
Glue	\$0.00	-	-	-	-	\$0.00	\$0.00
Key Management Service	\$0.00	-	-	-	-	\$0.00	\$0.00
Secrets Manager	\$0.00	-	-	-	-	\$0.00	\$0.00
Location Service	\$0.00	-	-	-	-	\$0.00	\$0.00
SNS	\$0.00	-	-	-	-	\$0.00	\$0.00
SQS	\$0.00	-	-	-	-	\$0.00	\$0.00
CloudShell	\$0.00	-	-	-	-	-	\$0.00
CloudWatch	\$0.00	-	-	-	-	-	\$0.00

This month's billing

Total active services: 22		Total pre-tax service charges in USD: USD 76.13
Description	Usage Quantity	Amount in USD
Relational Database Service		USD 48.37
Elastic Compute Cloud		USD 28.83
Virtual Private Cloud		USD 2.69
MQ		USD 2.12
WAF		USD 0.86
Key Management Service		USD 0.68
Route 53		USD 0.51
Secrets Manager		USD 0.07
API Gateway		USD 0.00
Certificate Manager		USD 0.00
CloudFormation		USD 0.00
Cloudfront Flat-Rate Plans		USD 0.00
CloudTrail		USD 0.00
CloudWatch		USD 0.00
Data Transfer		USD 0.00
Elastic Load Balancing		USD 0.00
Glue		USD 0.00
Lambda		USD 0.00
Location Service		USD 0.00

Cost Comparison & Insights (Month-over-Month)

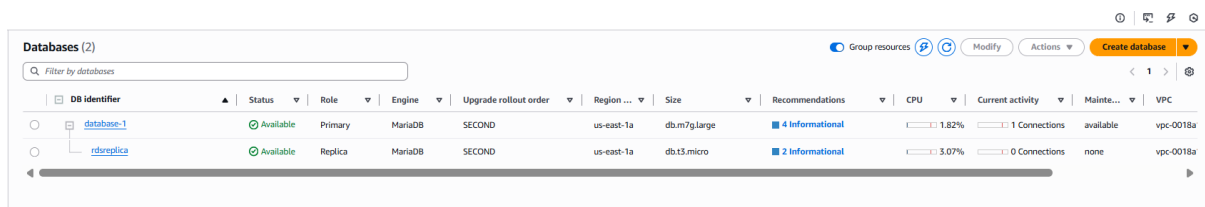
- The **current month AWS spend increased to ~\$76**, compared to **~\$25.5 in the previous month**, indicating temporary over-provisioning of resources.
- The **major cost contributors were RDS (~\$40) and EC2 (~\$28)**, which were identified as **unused or not required**.
- Additional minor costs came from **VPC, KMS, Secrets Manager, MQ, and tax**, contributing to operational overhead.
- After reviewing usage, **non-essential services were deleted or downsized**, including RDS, MQ, unused EC2 sizing, EBS volumes, backups, and Elastic IPs.
- These optimizations significantly **reduced recurring monthly costs**, bringing the environment back to a **lean, usage-based cost model**.

Key Optimization Outcome (Summary Line)

Month-over-month cost analysis helped identify idle and over-provisioned resources, enabling targeted clean-up actions that prevented continued unnecessary spending.

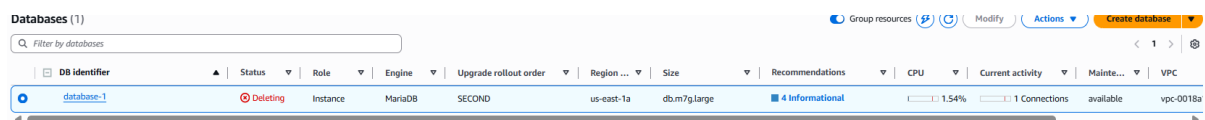
Step:-1

- The **RDS instance and read replica were not required** for the current workload.
- The database had **no data and no active application dependency**.
- Keeping an unused RDS instance was causing **unnecessary monthly charges (~\$45)**.
- By deleting the RDS and read replica, **wasted compute, storage, and backup costs were eliminated**.
- This action improved **cost efficiency and aligned resources with actual business needs**.



DB identifier	Status	Role	Engine	Upgrade rollout order	Region ...	Size	Recommendations	CPU	Current activity	Mainte...	VPC
database-1	Available	Primary	MariaDB	SECOND	us-east-1a	db.m7g.large	4 Informational	1.82%	1 Connections	available	vpc-0018a
rdsreplica	Available	Replica	MariaDB	SECOND	us-east-1a	db.m7g.large	2 Informational	3.07%	0 Connections	none	vpc-0018a

It was the RDS which is causing me above 45 dollars this month



DB identifier	Status	Role	Engine	Upgrade rollout order	Region ...	Size	Recommendations	CPU	Current activity	Mainte...	VPC
database-1	Deleting	Instance	MariaDB	SECOND	us-east-1a	db.m7g.large	4 Informational	1.54%	1 Connections	available	vpc-0018a

Deleted RDS snapshot

The screenshot shows the AWS RDS console. At the top, a green banner states "Successfully deleted snapshot database-1-snapshot." Below this, the "Snapshots" section is active, showing a table of manual snapshots. The table has columns for Snapshot name, Engine version, DB instance or cluster, Snapshot creation time, DB Instance created time, Status, Progress, and VPC. Three snapshots are listed: "database-1-snapshot" (status: Creating...), "rdabackup" (status: Available...), and "snap2" (status: Available...). Below the table, a message states "No Snapshots found, you can take a snapshot using Take Snapshot button".

Snapshot name	Engine version	DB instance or cluster	Snapshot creation time	DB Instance created time	Status	Progress	VPC
database-1-snapshot	11.8.5	database-1	December 22, 2025, 14:52 (UTC+05:30)	December 11, 2025, 20:18 (UTC+05:30)	Creating...	0%	vpc-0018a135ed108a41b
rdabackup	11.8.5	database-1	December 11, 2025, 20:03 (UTC+05:30)	December 11, 2025, 16:31 (UTC+05:30)	Available...	Completed	vpc-0018a135ed108a41b
snap2	8.0.43	database-2	December 11, 2025, 20:11 (UTC+05:30)	December 11, 2025, 19:56 (UTC+05:30)	Available...	Completed	vpc-0018a135ed108a41b

Step:-2

- The EC2 instance t3a.xlarge was over-provisioned for the actual workload.
- Resource utilization (CPU and memory) did not justify a large instance size.
- The instance was downsized to t2.micro, which meets current performance needs.
- This right-sizing reduced unnecessary compute capacity.
- The change results in approximately \$30/month cost savings while maintaining functionality.

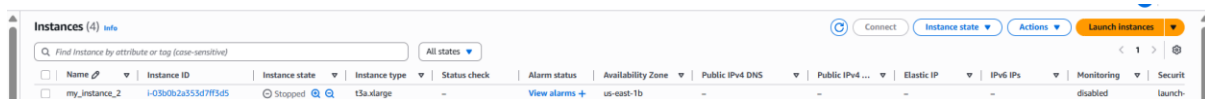
approximate comparison on a monthly basis (On-Demand, 24×7 running). Costs vary slightly by **region**, but this is good for documentation and explanation:

- **t3a.xlarge**
 - ~ **\$0.15–\$0.17 per hour**
 - ~ **\$110–\$125 per month**
 - 4 vCPU, 16 GB RAM (high capacity, higher cost)
- **t2.micro**
 - ~ **\$0.011–\$0.013 per hour**
 - ~ **\$8–\$10 per month**
 - 1 vCPU, 1 GB RAM (low usage, cost-efficient)

Monthly Difference

- **Savings: approximately \$100–\$115 per month**
- Achieved by right-sizing based on actual workload usage

I have 1 ec2 instance with t3a.xlarge I am going with right size t2micro



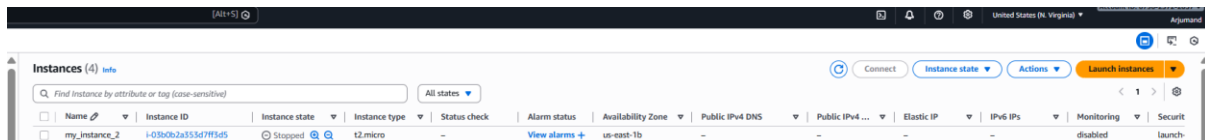
Instances (4) info

Find instance by attribute or tag (case-sensitive)

All states

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elastic IP	IPv6 IPs	Monitoring	Security
my_instance_2	i-03b0b2a353d7ff5d5	Stopped	t3a.xlarge	-	View alarms +	us-east-1b	-	-	-	-	disabled	launch-

Changed the instance to t2micro



Instances (4) info

Find instance by attribute or tag (case-sensitive)

All states

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elastic IP	IPv6 IPs	Monitoring	Security
my_instance_2	i-03b0b2a353d7ff5d5	Stopped	t2.micro	-	View alarms +	us-east-1b	-	-	-	-	disabled	launch-

Step: 3

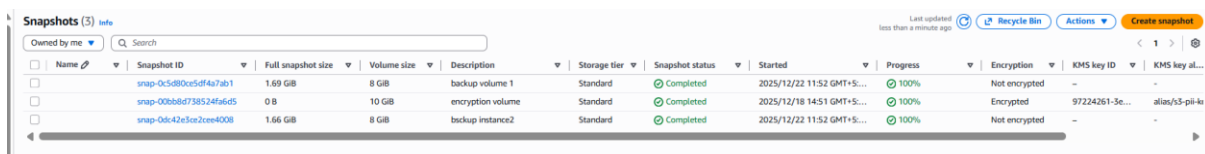
Deleting unwanted volume and backups

- Multiple unused EBS volumes and backups (snapshots) were identified with no active EC2 dependency.
- Keeping unattached volumes and old backups was causing unnecessary storage charges.
- 2 unused EBS volumes (~50 GB each) were costing approximately \$8–\$10 per month.
- 2 EBS snapshots (backups) were adding another \$4–\$6 per month in storage costs.
- Deleting these unused resources resulted in ~\$12–\$16 monthly savings and improved storage hygiene.

One-line summary (optional):

Removing unused EBS volumes and backups eliminated unnecessary storage costs while keeping only required resources.

I have 3 snaps which is not necessary and free to delete



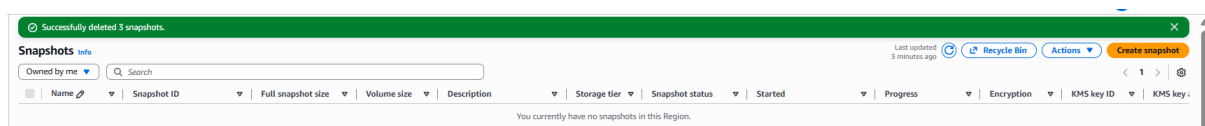
Snapshots (3) info

Owned by me

Search

Name	Snapshot ID	Full snapshot size	Volume size	Description	Storage tier	Snapshot status	Started	Progress	Encryption	KMS key ID	KMS key al...
	snap-0c5d80ce5df4a7ab1	1.69 GiB	8 GiB	backup volume 1	Standard	Completed	2025/12/22 11:52 GMT+5...	100%	Not encrypted	-	-
	snap-00bbbd738524fa6d5	0 B	10 GiB	encryption volume	Standard	Completed	2025/12/18 14:51 GMT+5...	100%	Encrypted	97224261-3e...	alias/s3-pii-k...
	snap-0d642e3ee2ce4008	1.66 GiB	8 GiB	backup instance2	Standard	Completed	2025/12/22 11:52 GMT+5...	100%	Not encrypted	-	-

Deleted snapshots which was not necessary



Successfully deleted 3 snapshots.

Snapshots info

Owned by me

Search

Name	Snapshot ID	Full snapshot size	Volume size	Description	Storage tier	Snapshot status	Started	Progress	Encryption	KMS key ID	KMS key
You currently have no snapshots in this Region.											

I have 6 volume which was not in use will save upto 6-8 dollars per month

Volumes (7) info

Last updated less than a minute ago [Recycle Bin](#) [Actions](#) [Create volume](#)

☐	Name	Volume ID	Type	Size	IOPS	Throughput	Snapshot ID	Source volume ID	Created	Availability Zone	Volume state	Alarm status	Attached resource
☐		vol-09093824a0b99ae4	gp3	100 GiB	3000	125	-	-	2025/12/22 11:53 GMT+5:30	use1-az1 (us-east-1a)	Available	No alarms	+
☐		vol-0f5b2c69dc7941cd	gp3	150 GiB	3000	125	-	-	2025/12/22 11:54 GMT+5:30	use1-az1 (us-east-1a)	Available	No alarms	+
☐		vol-019889c2261b1cf54	gp3	8 GiB	3000	125	snap-0434ec3...	-	2025/12/10 15:30 GMT+5:30	use1-az2 (us-east-1b)	Available	No alarms	+
☐		vol-0aa4ac341b89e7f0	gp3	8 GiB	3000	125	snap-0434ec3...	-	2025/12/20 17:02 GMT+5:30	use1-az2 (us-east-1b)	In-use	No alarms	+
☐		vol-032896096a525cb01	gp3	8 GiB	3000	125	snap-066759f...	-	2025/12/18 17:26 GMT+5:30	use1-az1 (us-east-1a)	Available	No alarms	+
☐		vol-01ff02d0fb3ba956	gp3	10 GiB	3000	125	-	-	2025/12/22 11:55 GMT+5:30	use1-az1 (us-east-1a)	Available	No alarms	+
☐		vol-086c889e596626dc8	gp3	10 GiB	3000	125	-	-	2025/12/18 14:39 GMT+5:30	use1-az1 (us-east-1a)	Available	No alarms	+

Snapshot summary
Recently backed up volumes / Total # volumes
0 / 7

Data Lifecycle Manager default policy for EBS Snapshots status
No default policy set up | [Create policy](#)

Last updated on Mon, Dec 22, 2025, 03:27:50 PM (GMT+05:30)

Deleted volumes which is not required

Volumes (1) info

Last updated less than a minute ago [Recycle Bin](#) [Actions](#) [Create volume](#)

☐	Name	Volume ID	Type	Size	IOPS	Throughput	Snapshot ID	Source volume ID	Created	Availability Zone	Volume state	Alarm status	Attached resource
☐		vol-0aa4ac341b89e7f0	gp3	8 GiB	3000	125	snap-0434ec3...	-	2025/12/20 17:02 GMT+5:30	use1-az2 (us-east-1b)	In-use	No alarms	+

Steps: 4

- Identified unused VPCs that had no active workloads or dependencies.
- Retaining unused VPCs increased management complexity without any business value.
- Unassociated Elastic IPs were incurring hourly charges while not being used.
- Each unused Elastic IP costs approximately \$3–\$4 per month.
- Deleting unused VPCs and releasing Elastic IPs resulted in direct cost savings and a cleaner network architecture.

Step: 4

Releasing unnecessary IPs and deleting KMS keys

Elastic Ip is not required for me as of now, if I want, I will keep it or else we can go ahead and delete, if we are working in any organisation will seek assistance/permission before deleting any AWS resources

Elastic IP addresses (1/2) info

Find elastic IP addresses by attribute or tag [Actions](#) [Allocate Elastic IP address](#)

☐	Name	Allocated IPv4 address	Type	Allocation ID	Reverse DNS record	Associated instance ID	Private IP address	Association ID	Network interface on
☒	-	34.192.97.102	Public IP	eipalloc-0277e0e85d38fdb4b	-	-	12.0.5.250	eipassoc-0ea9a89d2e1b1ae2e	679625722057
☐	-	52.22.110.74	Public IP	eipalloc-061caa3d179b9f59	-	-	12.0.0.15	eipassoc-041ca5225b89068c4	679625722057

Deleted nat gateway

NAT gateways (1/1) info

Find NAT gateways by attribute or tag [Actions](#) [Create NAT gateway](#)

☒	Name	NAT gateway ID	Connectivity	State	State message	Availability	Route table ID	Primary public IP	Primary private IP	Primary network interface	VPC
☒	mynatgateway	nat-0bc7180635f940d6	Public	Available	-	Zone1	-	52.22.110.74	12.0.0.15	eni-0d5bde0654b7...	vpc-0d1bda115eaf10841b / del...

Delete NAT gateway

Will be deleted
The following NAT gateway will be deleted permanently and can't be recovered later.

Name	NAT gateway ID	State
mynatgateway	nat-0bc7180635f940d6	Available

To confirm deletion, type delete in the field:

[Cancel](#) [Delete](#)

nat-0bc7180635f940d6 / mynatgateway

[Details](#) [Secondary IP addresses](#) [Monitoring](#) [Tags](#)

Created KMS keys for practice purpose, no need of keys to be stoped

Customer managed keys (2)

Filter keys by properties or tags

Aliases	Key ID	Status	Key type	Key spec	Key usage
alias/3-pii-key	97224261-3e1d-4d5a-8c8a-637b88a5f037	Enabled	Symmetric	SYMMETRIC_DEFAULT	Encrypt and decrypt
my-key	9c06a66d-13ab-4189-8d00-fc34cabbd764	Enabled	Symmetric	SYMMETRIC_DEFAULT	Encrypt and decrypt

Schedule for deleting

Successfully scheduled deletion for 2 keys

Notifications 0 0 2 0 0 0

Customer managed keys (2)

Filter keys by properties or tags

Aliases	Key ID	Status	Key type	Key spec	Key usage
alias/3-pii-key	97224261-3e1d-4d5a-8c8a-637...	Pending deletion	Symmetric	SYMMETRIC_DEFAULT	Encrypt and decrypt
my-key	9c06a66d-13ab-4189-8d00-fc3...	Pending deletion	Symmetric	SYMMETRIC_DEFAULT	Encrypt and decrypt

Deleting Secrete Keys

Secrets

Filter secrets by name, description, tag key, tag value, owning service or primary Region

Secret name	Description	Last retrieved (UTC)	Created on (UTC)
prod/api/thirdparty/keys	Production third-party API keys	December 18, 2025	December 18, 2025 at 11:57:12
prod/rh/ingred/credentials	Production MySQL database credentials	December 18, 2025	December 18, 2025 at 11:09:23

Secrets

Filter secrets by name, description, tag key, tag value, owning service or primary Region

Secret name	Description	Last retrieved (UTC)	Created on (UTC)
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Deleting parameter store

Introducing cross-account parameter sharing

You can now use AWS Resource Access Manager to share parameters with other accounts in your organization. [Learn more](#)

Share in IAM

My parameters

Search

Name	Tier	Type	Last modified
/prod/app/lib_host	Standard	String	Fri, 19 Dec 2025 09:17:54 GMT
/prod/app/lib_password	Standard	SecureString	Fri, 19 Dec 2025 12:53:20 GMT

Deleted parameter store which was not required

Your parameters were successfully deleted

My parameters

Search

Name	Tier	Type	Last modified
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You do not have any parameters in this region.

Conclusion on KMS, Secret management key & system parameter store:

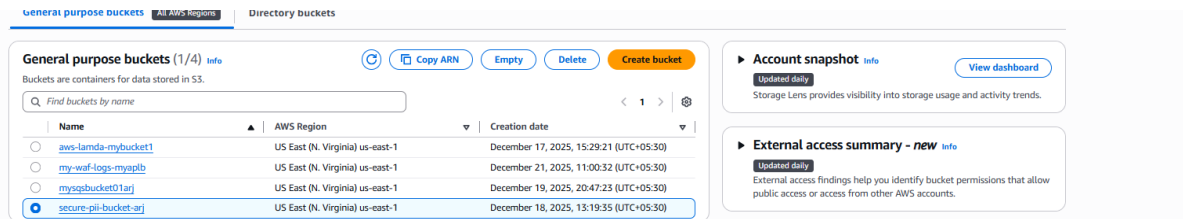
- 2 unused KMS keys were deleted, saving approximately \$2 per month (\$1 per key).
- 2 unused Secrets Manager secrets were removed, saving about \$0.80 per month (\$0.40 per secret).

- 2 Parameter Store entries had no additional cost, but deleting them reduced security and management overhead, with total savings of ~\$2.8 per month.

Step: 6

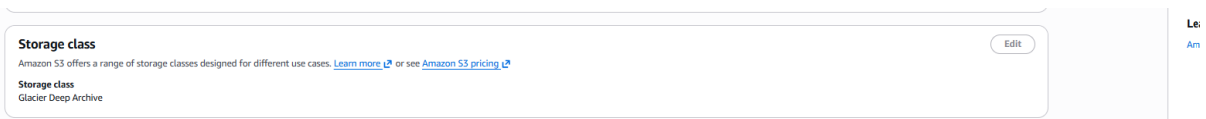
Deleting unwanted buckets and selecting right bucket storage

I will delete 3 buckets and change 1 bucket storage standard to glacier deep archive



Storage type change to deep archive

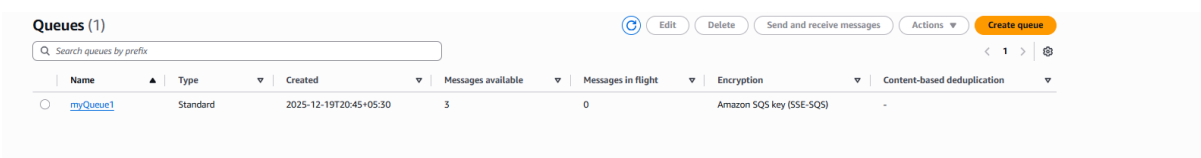
- **S3 Standard:** ~\$0.023 per GB/month – designed for frequent access.
- **S3 Glacier Deep Archive:** ~\$0.00099 per GB/month – for long-term archival with rare access.
- **Savings:** Deep Archive is **about 95–96% cheaper** than Standard, ideal for backups and compliance data.



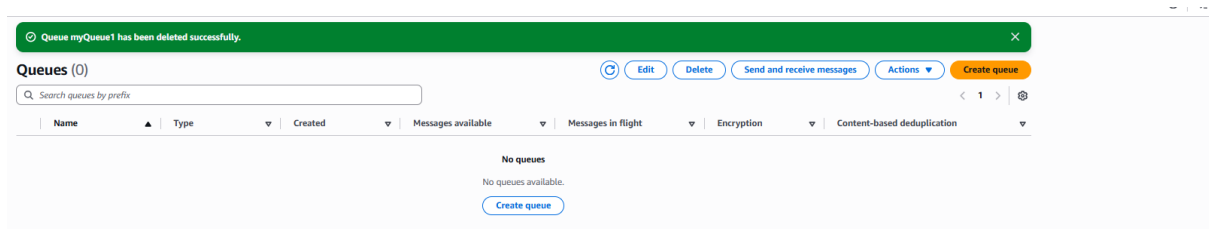
Step:7 Deleting SQS and Amazon MQ

- Amazon SQS has no fixed monthly cost; since the queue was unused, deleting it avoids future request and data transfer charges (usually negligible but good hygiene).
- Amazon MQ broker has a fixed hourly cost; even a small broker typically costs ~\$40–\$50 per month when running idle.
- Deleting the unused SQS and MQ results in ~\$40–\$50 monthly savings, mainly from removing the unnecessary MQ broker.

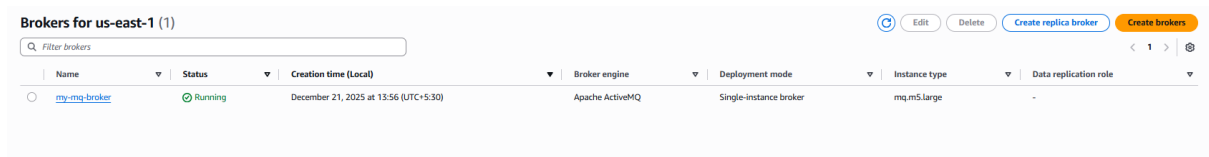
Deleting SQS, MQ that was not necessary



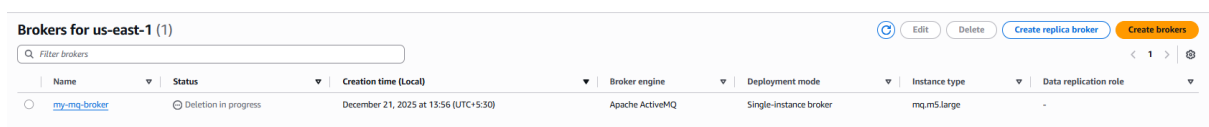
Deleted SQS



Deleting Amazon MQ



Deleted Amazon MQ it was not necessary if you we want we can keep



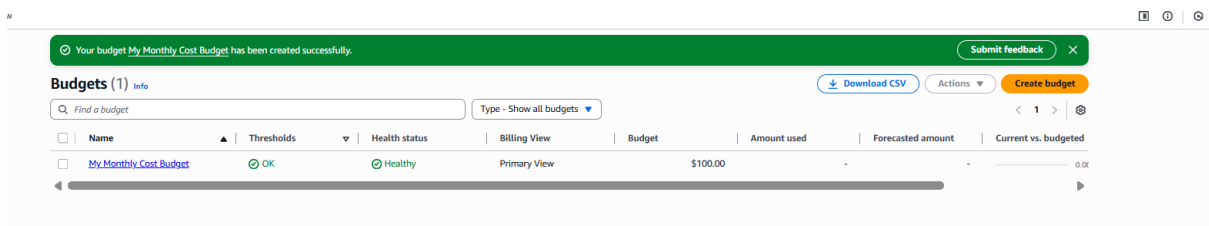
Step : 8

Creating budget which will keep you update if cost goes above the Threshold

Importance of budget planning:

- Setting a budget alert email at \$100 helps detect unexpected cost spikes early.
- It enables proactive action before bills exceed the planned budget.
- Budget alerts promote cost accountability and financial discipline in cloud usage.

Created budget which given \$100 as thersold value



While creating budget planner

Regular cost analysis and right-sizing directly translate into lower AWS bills, better governance, and a more efficient cloud environment.

